

Question Paper

Exam Date & Time: 24-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Augmented and Virtual Reality [CSE_4051]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain with a suitable example how the Three I's of Virtual Reality guide the design of a VR-based virtual classroom for interactive learning. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) A company is developing an immersive temple exploration experience using AR/VR technologies. (5)
- i) Analyze the key characteristics of VR that enhance virtual heritage experiences.
- ii) Discuss how AR can complement VR in providing contextual cultural information.
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) A global company is designing a VR workspace where users perform static, dynamic, symbolic, and manipulative gestures to interact. (5)
- A) Apply suitable techniques for real-time gesture input mapping and synchronization of multiple users within the shared virtual environment.
- Course Outcome: "CO2" / Cognitive Level: "Application"
- B) A smart city AR navigation app guides pedestrians using arrows on the screen, voice directions, and subtle vibrations when taking turns. (5)
- i) Analyze how visual and auditory feedback together help users follow the correct route.
- ii) Analyze how haptic cues improve user safety and attention in busy environments.
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) A VR archaeological exploration app allows users to walk through a recreated ancient city using full-body tracking and multimodal feedback. Apply your understanding of AR/VR interface technologies to improve immersion. (5)
- A) i) Explain how 3D position trackers can help users navigate archaeological sites accurately.
- ii) Apply binaural and ambisonic audio techniques to guide users to hidden artifacts or important locations.
- iii) Illustrate the use of tactile feedback when users virtually touch artifacts.
- iv) Discuss how gesture-based manipulation can allow object inspection from different angles.
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) A VR art museum lets users experience paintings and sculptures through multiple sensory outputs. (5)
- i) Analyze how geometric modelling enhance the visual realism of paintings and sculptures.
- ii) Examine how 3D spatial sound contributes to immersion while exploring the museum environment.

- iii) Discuss the role of haptic feedback and behavior/kinematic modelling in making sculpture interaction feel authentic.
- iv) Evaluate challenges in synchronizing visual, audio, haptics feedback without overwhelming the user.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) A mobile AR device uses incremental tracking while performing SLAM during user movement. (5)
- A) i) Apply the motion model concept to explain how predicting the next feature location helps reduce computation during tracking.
- ii) Apply the combined detection and tracking approach to explain how the device maintains reliable feature points when some are lost from view. Draw a simple block diagram showing how detection and tracking work together in the process.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Distributed VR systems use side-by-side displays and PC clusters to support multi-user immersive environments. (5)
- i) Analyze how synchronization methods (software, frame-buffer, and video synchronization) impact performance and visual consistency in these architectures.
- ii) Support your answer with a simplified diagram of synchronized rendering pipelines.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) A robotic arm model in an AR-based manufacturing simulator must position its gripper accurately to pick up tools from a workbench. (5)
- A) Apply kinematic modeling by describing:
- i) The application of kinematic chain elements (links, joints, actuators) in determining the robot's reachable workspace.
- ii) The application of numerical inverse kinematics in updating joint angles when the target tool moves slightly, while avoiding unnatural or impossible poses.

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) In a collaborative VR design platform, multiple users can modify the same 3D model of a car at the same time. Some users report inconsistencies, changes made on one headset do not instantly appear on others. (5)
- i) Analyze the need for model synchronization in multi-user VR and its impact on interactivity.
- ii) Explain how model hierarchy or scene graph can help maintain consistency during real-time updates.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_4051:Augmented and Virtual Reality

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	4	20(40%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)